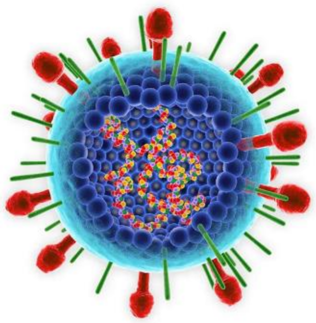
# The Belgian National Survey on RSV pediatric burden of disease



Preliminary results

Belgian Academy Congress, March 2025

Blumental S, De Mot L, Raes M

and **the Belgian Pediatric RSV Study Group and the BELSARI-net research group**


# Setting


- 24

→ Creation of the RSVPED study, run in parallel to SARI survey, bringing complementary pediatric data

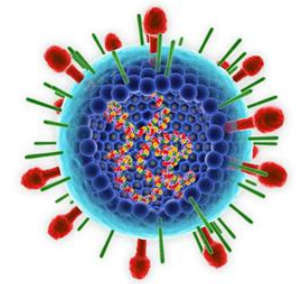
- infections in

+ **Update of SARI protocol** to better catch pediatric cases (from November 2023)

- **→Pooled analysis of both datasets at the end**

Decision to build **a broad surveillance network** to better **assess the RSV pediatric burden of disease in Belgium (first in hospitals setting)** and to **monitor impact of preventive tools** once implemented

# Method

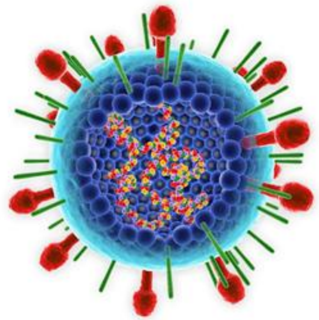


- **New common definition for pediatric severe acute respiratory infections** and a **new questionnaire** adapted to pediatric features, almost similar between **both studies**

- **SARI survey**
  - Prospective inclusion of all cases hospitalized for severe acute respiratory infections matching the case definition
  - Sample sent to the reference Lab, identification of respiratory viruses by pcr
  - Weekly clinical data report by participating centers (pseudonymized data)
  - Weekly bulletin of acute respiratory infections

- **RSVPED study**
  - Active recruitment of all centers in Belgium outside the SARI network
  - Inclusion of **all proven RSV** cases **0-4 years old** meeting the case definition and hospitalized during the enlarged epidemic period (October to March)
  - Retrospective **clinical data collection** by local team/data nurse (short or long CRF) (pseudonymized data for season 2023-2024)
  - Inclusion of **possible cases** for centers not testing all children hospitalized with bronchiolitis
  - Preliminary aggregated data obtained for the 3 first months of the current season

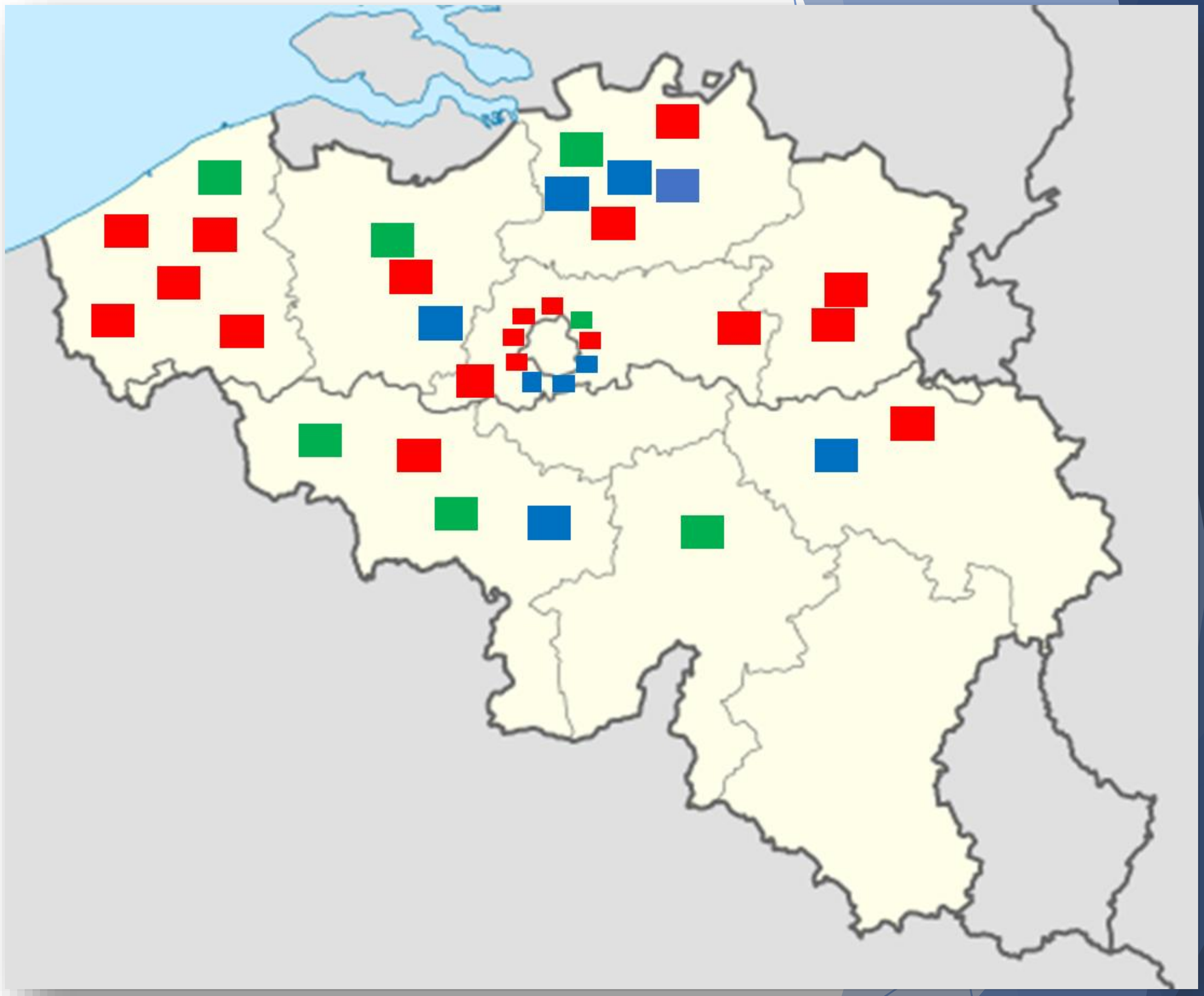
# Timing



- **3 years of surveillance**
  - 2023-2024 (pre-prevention period);
  - 2024-2025 (first year of prevention implementation, one tool reimbursed),
  - Future 2025-2026 (second year of prevention, two tools reimbursed)

- **Merged of both datasets after each season and pooled analysis for common items**
- **Specific analyses leveraging main assets of each study**

# 34 participating sites all over the country



- RSVPED : 27 centers
  - 19 centers from which incidence data from both seasons are available (paired data)
  - 8 centers partial data collection
  - And many more coming…!!
- SARI : 7 additional centers

UZ Brussels, UZ Gent, AZ Sint Jan Brugges, Centre Chwapi Tournai, CHU Namur-Godinnes, Grand Hôpital de Charleroi, ZAS ziekenhuizen, CHU St Pierre, CHU Tivoli, AZ Klina, AZ Turnhout, AZ Delta Roeselare, Jessa Ziekenhuis, Clinique St jean, AZ St Maria Halle, AZ Jan Palfijn Gent, AZ Groeninge, Jan Yperman, Hôpital Delta Chirec, Hôpital St Anne-St Remy Chirec, Hôptial de Braine-l'alleul, HUDERF, OLV Wareghem, CHC Mont Leggia, AZ Diest, Imelda Ziekenhuis, Hôpital Iris Sud Ixelles, Vitaz, Cliniques de L'Europe, Cliniques Universitaires St Luc, Universiteit Ziekenhuis Antwerp, CHU/Citadelle Liège, AZ Blasius, CHU Charleroi

# RSV: high pediatric burden of disease season 2023-2024

- RSVPED
  - **2741 cases** included from **21 centers** (even under estimated!)
  - 54,5% < 6 months old, 91% < 2 years old
  - Median length of stay :5,4 days
  - Use of respiratory support in 61% of the cohort (67% in infants<6m)
  - Antibiotics consumption in 34% of children (26% in infants <6months)
  - 218 ICU admissions recorded from 5 PICU tertiary centers

# Based on Belgian epidemiology in 2023-2024 (SARI): how many **hospitalizations** could **hypothetically be prevented** if eligible patients receive **Nirsevimab**?

**Extrapolation of the benefits expectable based on:**

- The estimated incidence for RSV hospitalizations in children 0-4y in 2023-2024 :
  **1049.8 per 100.000 inhabitants <5y old**

- The proportion of admitted children that would have been eligible for immunization among all hospitalized cases (before prevention available)
  **66%** (newborns 29%; catch up 37%).

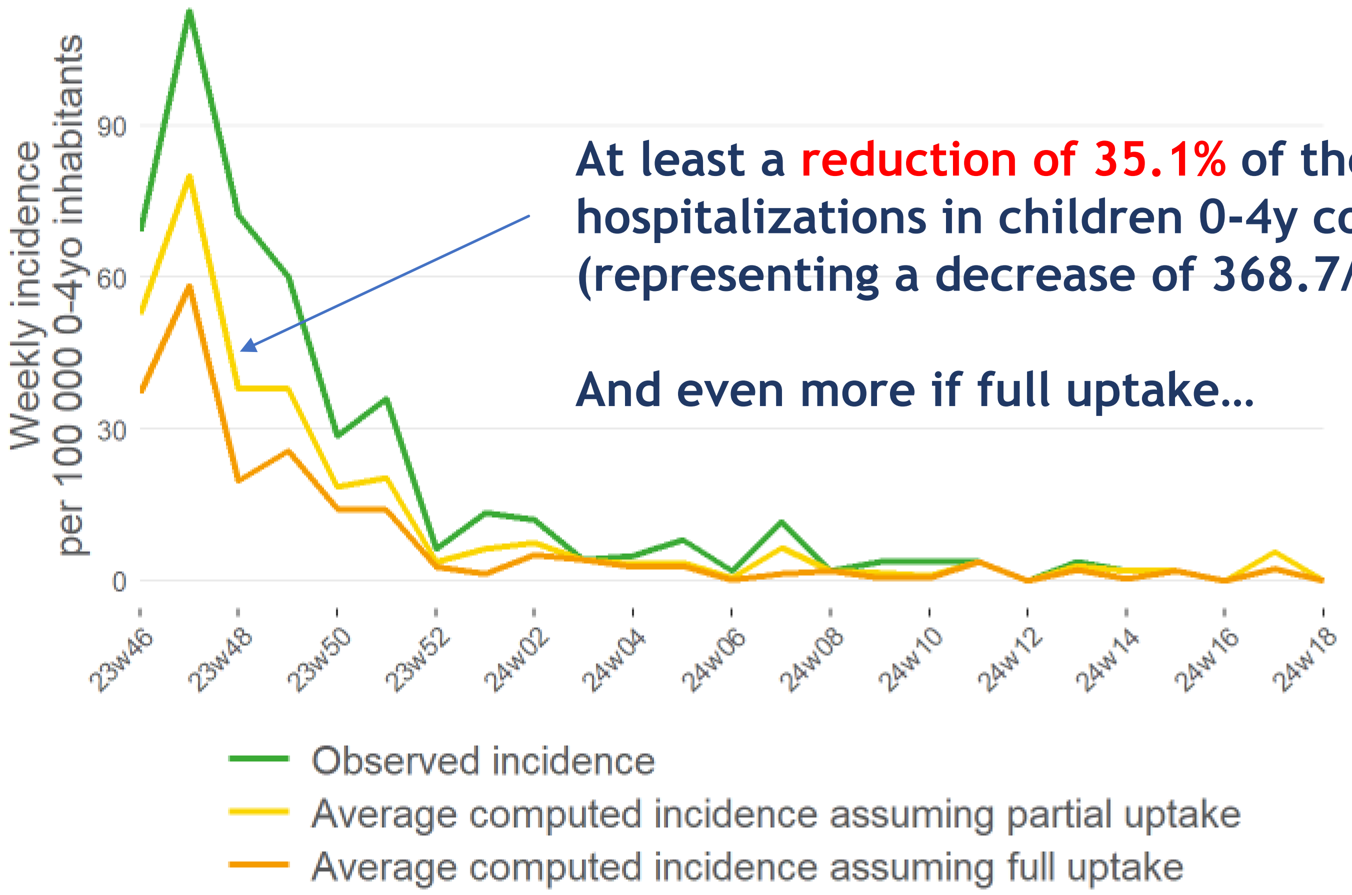
- The percentage of uptake of the preventive tool: ?? **85% maternity, 50% catch up**
- The efficacy of the preventive tool to prevent hospitalization (literature data) : **88%**

| Uptake | Newborn | Catch-up |
|---|---|---|
| Low | 0.75 | 0.3 |
| Medium | 0.85 | 0.5 |
| High | 0.9 | 0.6 |

x

| Efficacy | |
|---|---|
| Lower bound 95% CI | 0.85 |
| Average efficacy | 0.88 |
| Upper bound 95% CI | 0.91 |

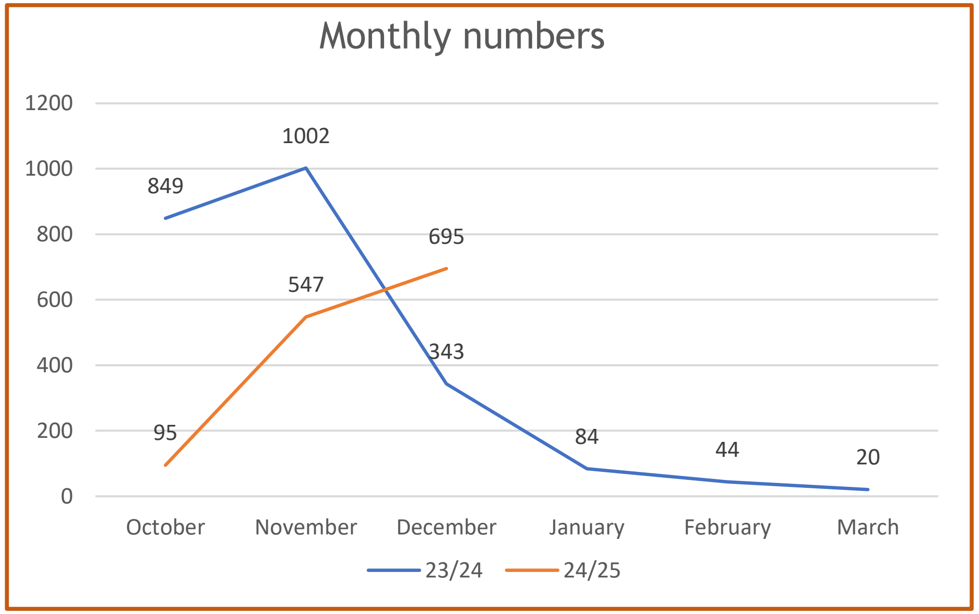
# Based on Belgian epidemiology in 2023-2024, **hypothetical impact** of **Nirsevimab** on pediatric hospitalization using **a medium uptake** and **average efficacy**

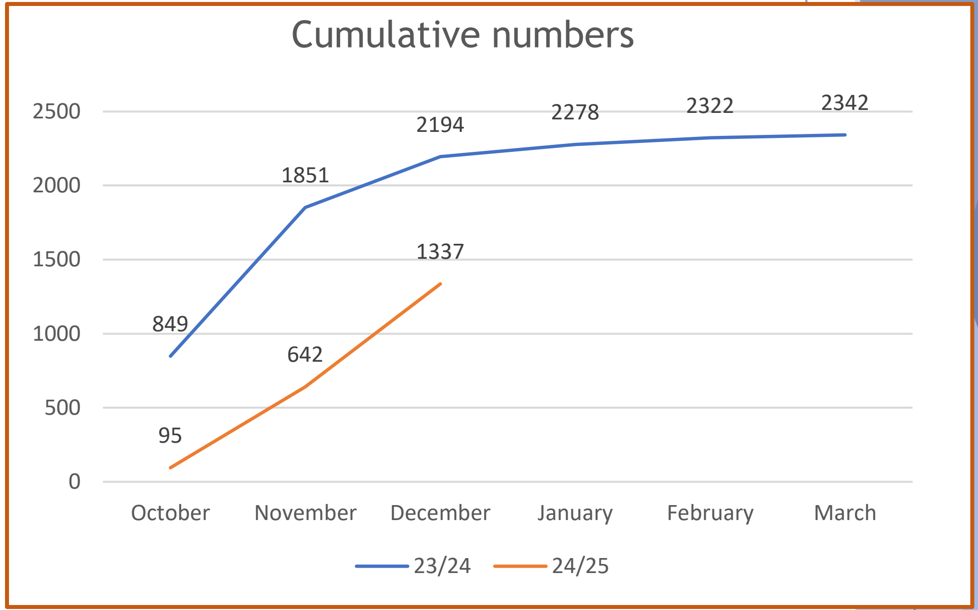



**At least a reduction of 35.1% of the total incidence of RSV hospitalizations in children 0-4y could be expected ! (representing a decrease of 368.7/100.000 children 0-4 y.)**

**And even more if full uptake…**

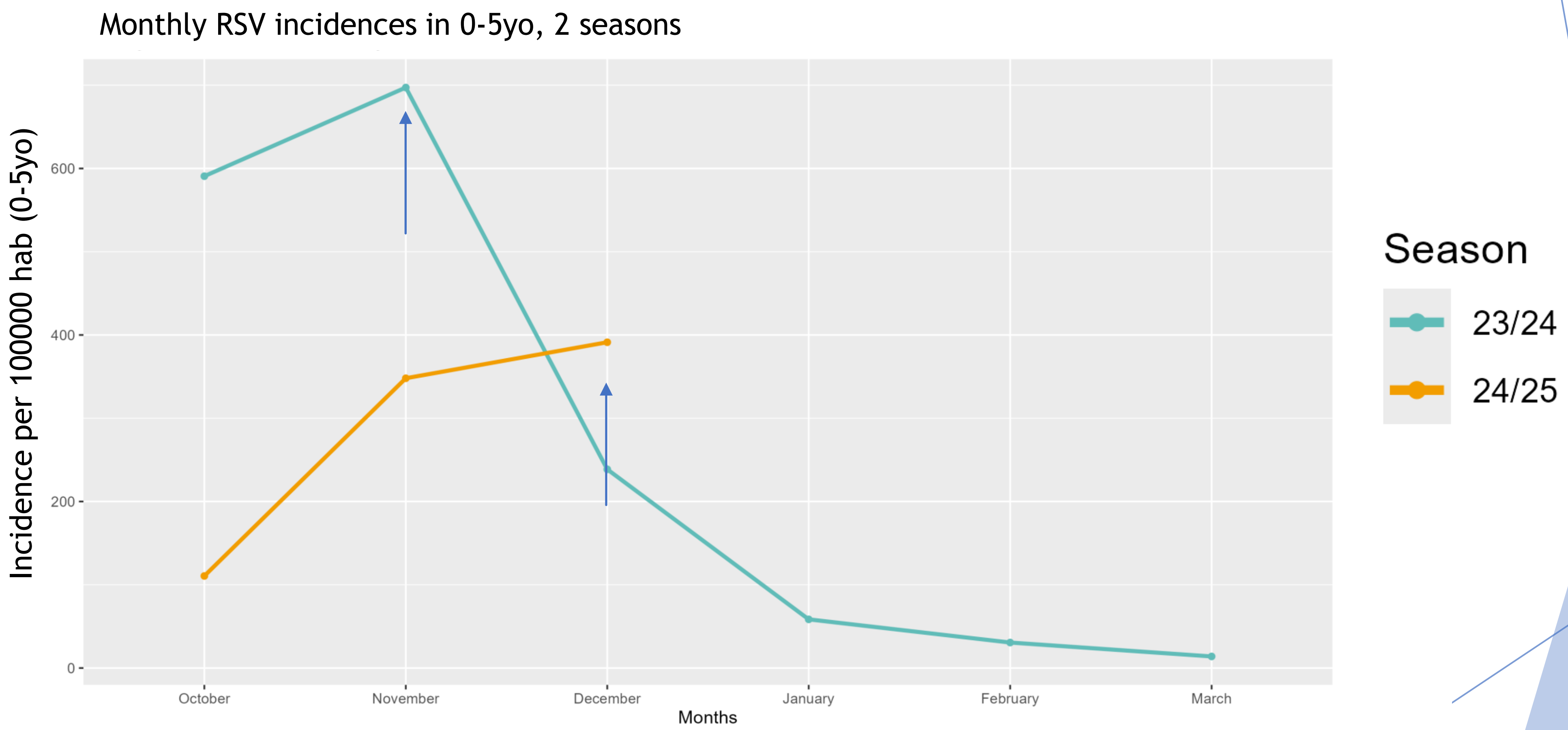
# Number of children 0-4 y old hospitalized for RSV in 2024-2025 (first year prevention) compared to 2023-2024 (RSVPED)



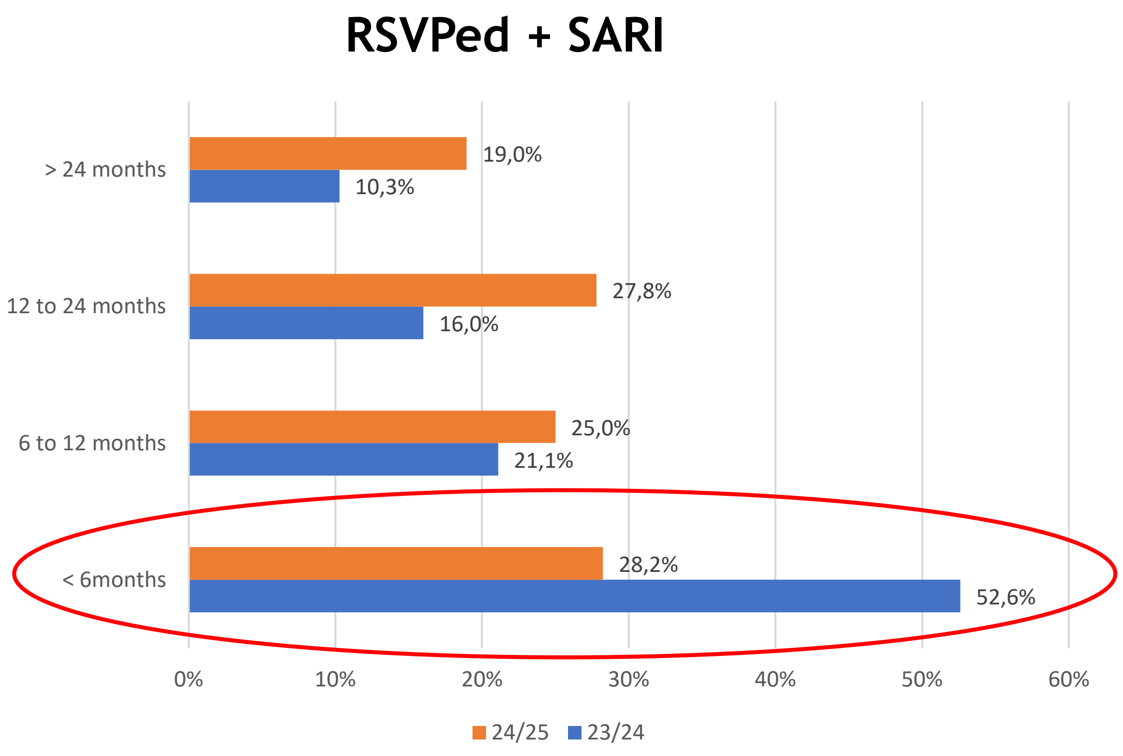

- Between October and December: **much less hospitalized cases** this season compared to the previous one (! Actual season ongoing)
- **Shift in seasonality**: **last season started earlier** than in the pre-pandemic period (start in September in 2023, October in 2024)

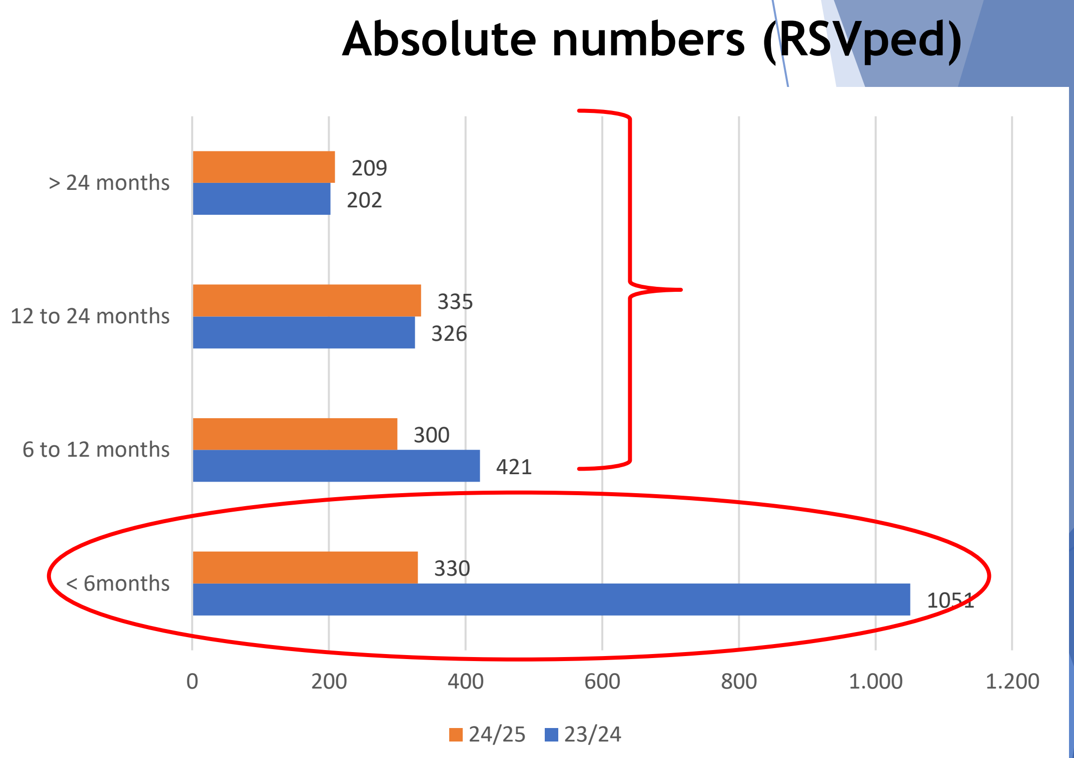
# Incidence patterns of RSV hospitalizations in children 0-4 y old in 2024-2025 and 2023-2024 (RSVPED)




- **Incidence at peak level**: ~400/100000 inhabitants 0-4y in 2024-25 versus ~700/100000 inhabitants 0-4y in 2023-24

# Age distribution among RSV pediatric hospitalized cases during both seasons (RSVPED and SARI)



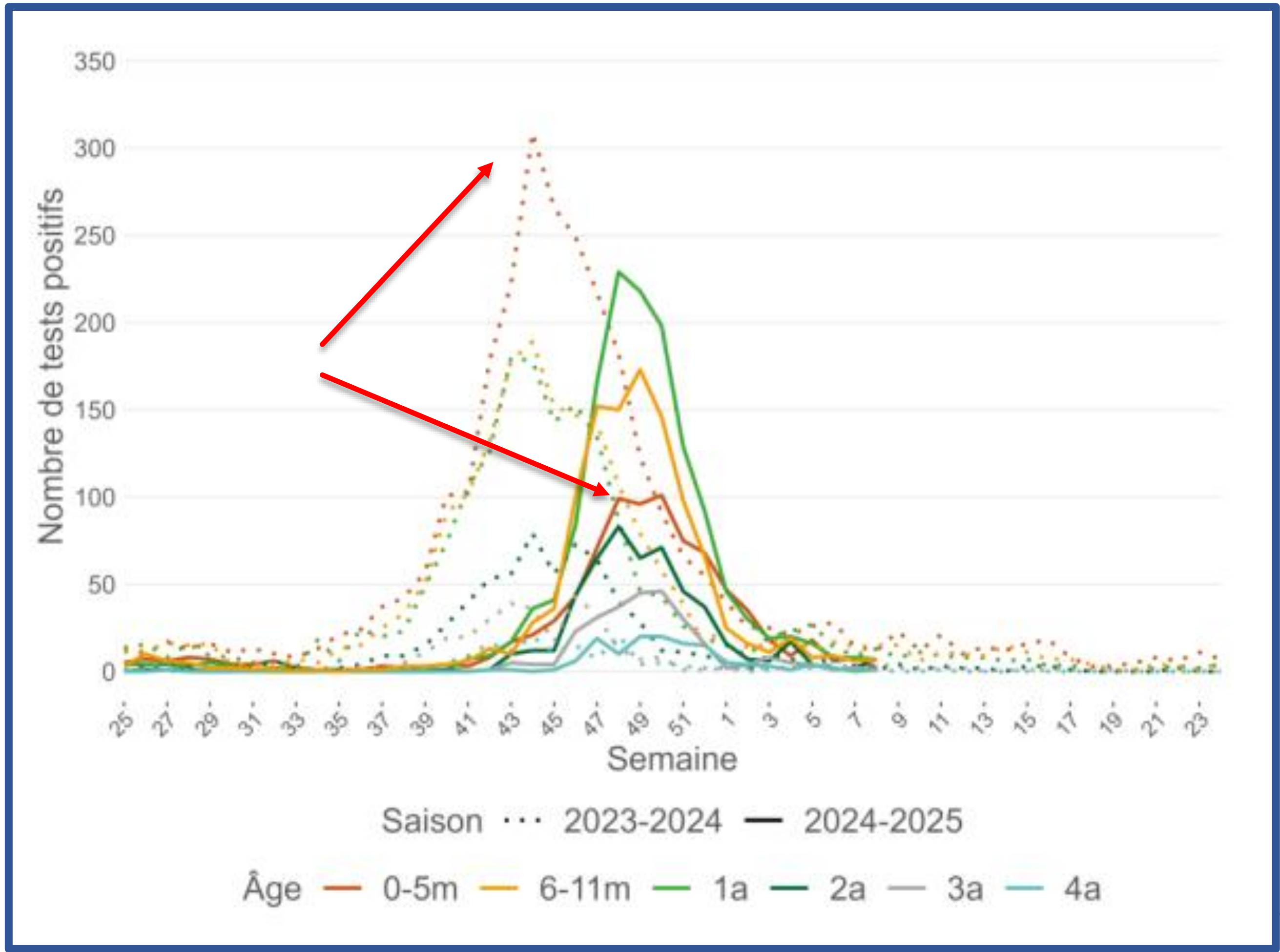




Decrease of burden in <6 months in 2024-2025 (proportionally less represented)

Concordant results between both studies

No changes in other age groups (no age shift)

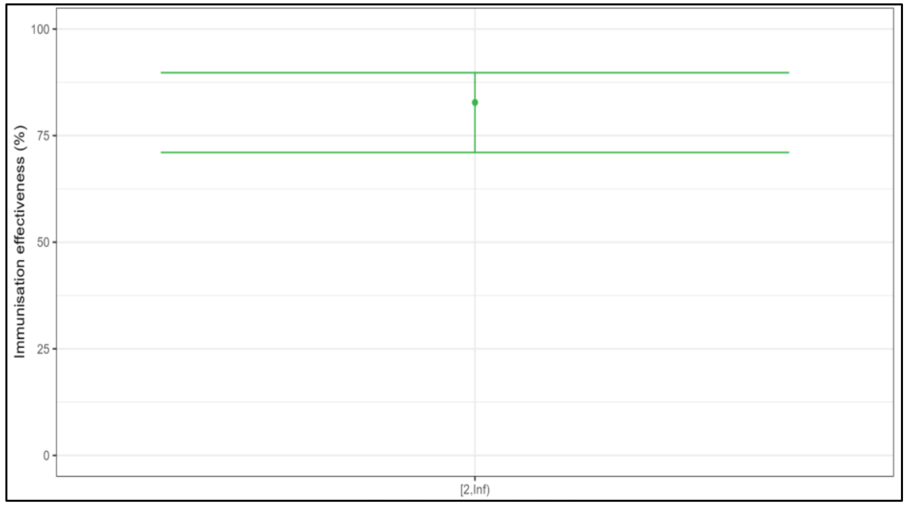
# Incidence pattern of ambulatory RSV cases in children 0-4y in 2023-2024 and 2024-2025
(sentinel labs, Sciensano)



# Effectiveness of nirsevimab in preventing hospitalization (SARI, 2024-2025)

Test-negative design (common in vaccine studies):

- Compares nirsevimab-eligible SARI patients hospitalized because of RSV (**cases**) to those hospitalized for SARI but tested negative for RSV (**controls**).
- If nirsevimab is effective, fewer immunized infants should be found among RSV-positive cases than among RSV-negative controls
- Adjusting for confounders (through logistic regression): comorbidities, sex, sampling week, age in months
- Excluding nosocomial infections, children who received palivizumab or whose mother received Abrysvo

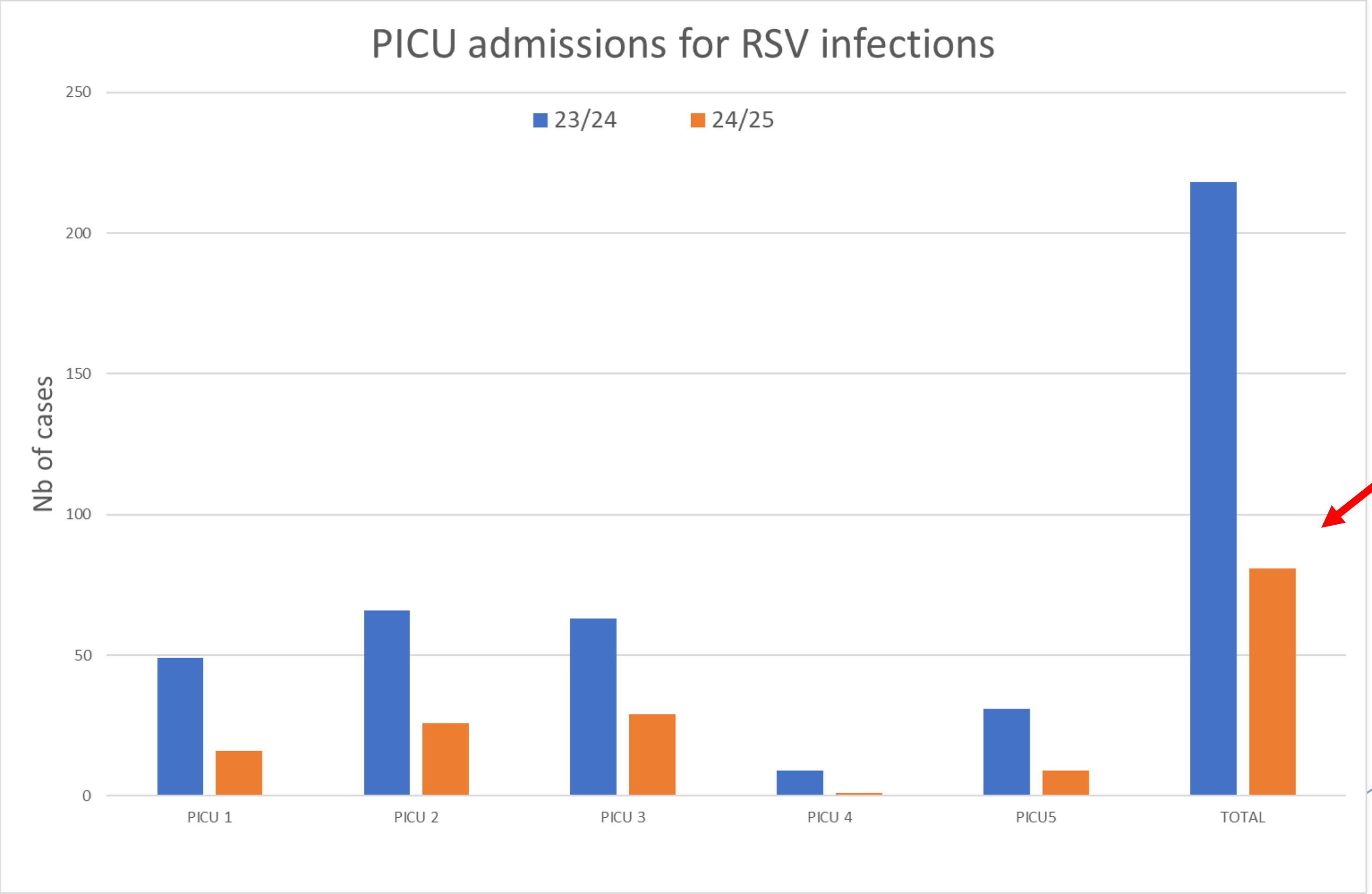



Immunization effectiveness = 83%

| | RSV-negative SARI hosp | RSV-positive SARI hosp |
|---|---|---|
| Not immunized | 111 | 170 |
| Immunized | 107 | 36 |

# Severity of infections/burden of disease ICU admissions (both studies)

- Decrease of total ICU admissions (5 main PICU centers, 3 regions)




# Immunization with Nirsevimab among hospitalized infants < 6 months (season 2024-2025)

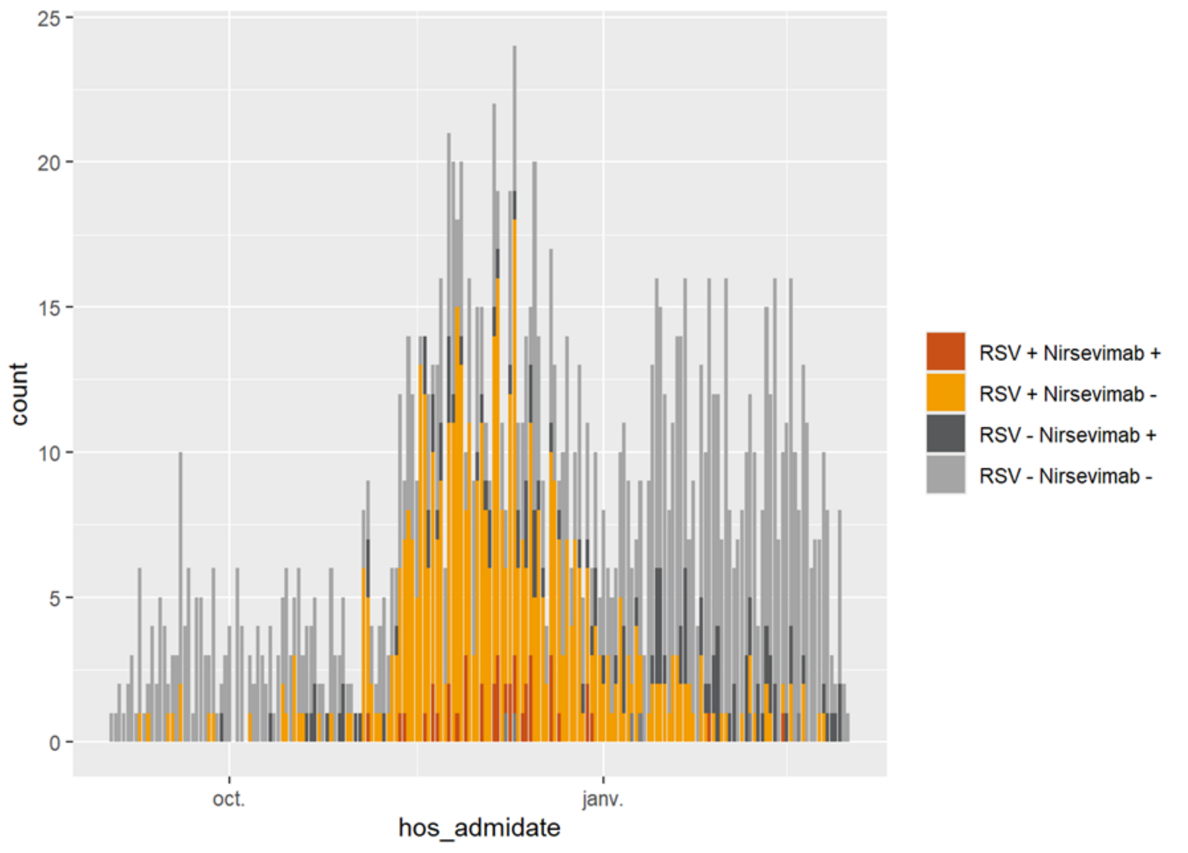
- 37/140 cases in SARI (**26%**)

(17 catch up program, 22 newborns)

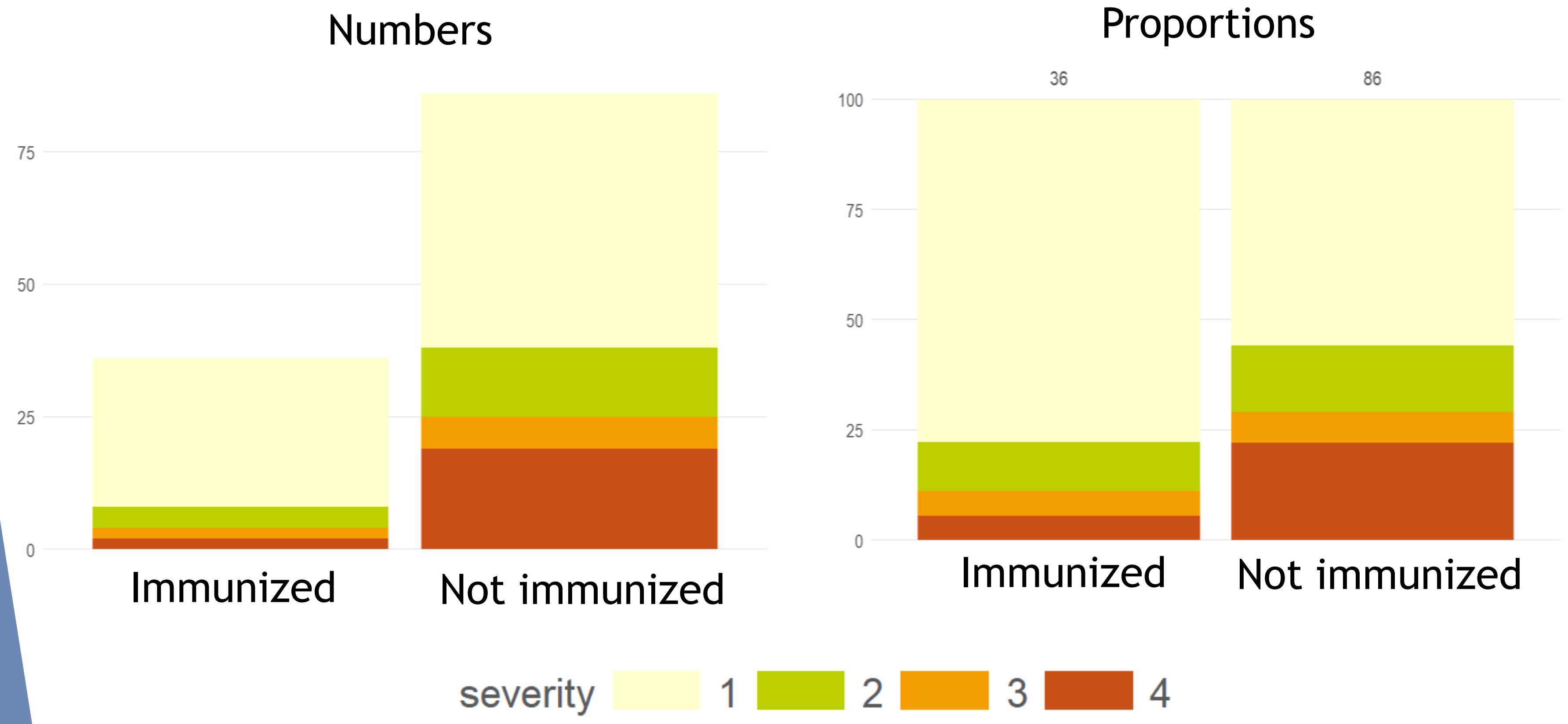
- 102/366 cases in RSVPed (**27,8%**)

(+ 9 with maternal vaccine)

- **Infants hospitalized and immunized with nirsevimab** :
  - represents 6 to 8% of the whole cohort of RSV hospitalized children 0-4y
  - has no more comorbidity (5% vs 9% in all eligible))
  - has no more prematurity (1% vs 5,5% in all eligible



# Severity of infections in infants eligible for Nirsevimab according to immunization status
(SARI, 2024-2025)




**4 grades of severity according to use of respiratory support**

**(1) = none**

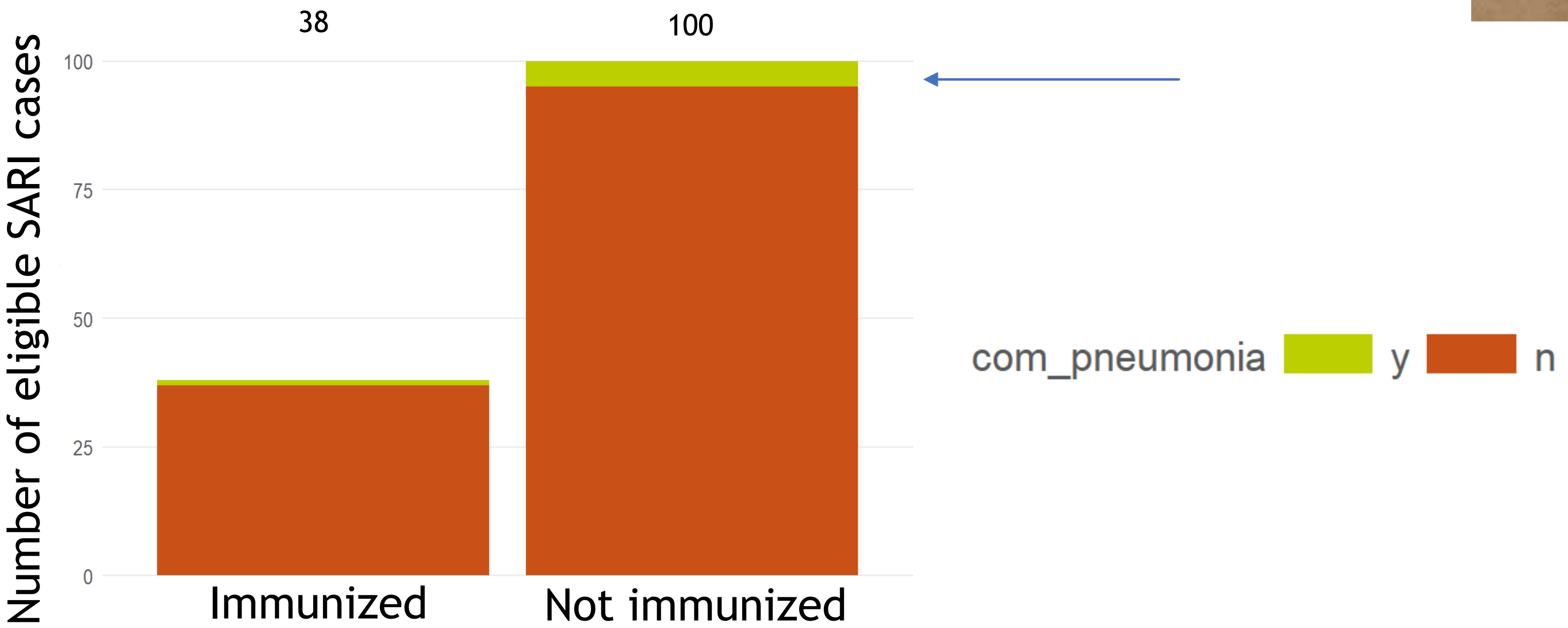
**(2) = oxygen goggles**

**(3) = mid-severity:** non-invasive ventilation

**(4) = highest severity:** invasive ventilation and ICU admission

**Infants immunized although hospitalized showed less severe infections!**

# Severity of infections in infants eligible for Nirsevimab according to immunization status
Rate of pneumonia (SARI, 2024-2025).




**To be analyzed yet to assess severity pattern of infection:**
length of stay, use of antibiotics, need for nutritional support, rate of neurological complications, total nb of transfers to ICU, details of ICU stays…

**+ Severity in older age groups (> 6m)**

What was the immunization uptake for this first year of prevention?

# Nirsevimab implementation in 30 hospitals
# Pilot Study (Pr Raes)

**1) Maternity** (25 hospitals from Flanders, Oct-Dec, same trend in Jan/Feb)

| | Octobre | Novembre | Decembre |
|---|---|---|---|
| Naissances | 3.546 | 2.755 | 2.714 |
| Couverture | (62, 12%) – **86%** – (96.82%) | (71.52%) – **81,75%** - (100%) | (82.29%) – **84,25%** – (100%) |

**2) Catch-up** (Octobre): 27 hospitals from Flanders/Brussels/Wallonia

9.580 inj / 27 hospitals → 35.500/100 hôpitaux/60.000 enfants/ 6 mois = **59%**

Injections/ invitations envoyées: 9.539/17.762 (**53,70%**)

+ **More data expected about ambulatory delivery** (intego, farmanet) **and national database** (IMA)

# If all eligible patients would have received Nirsevimab this season (100% uptake) : how many extra hospitalizations could have been prevented? (Oct 2024-Janv 2025, SARI data)

| immunisation_group | Nirsevimab | No_Nirsevimab | total_numb | irsevimab |
|---|---|---|---|---|
| All patients | 32 | 113 | | 22.06897 |
| catch-up | 14 | 72 | 86 | 16.27907 |
| newborn | 18 | 41 | 59 | 30.50847 |

**¾ of infants <6m hospitalized non immunized..**

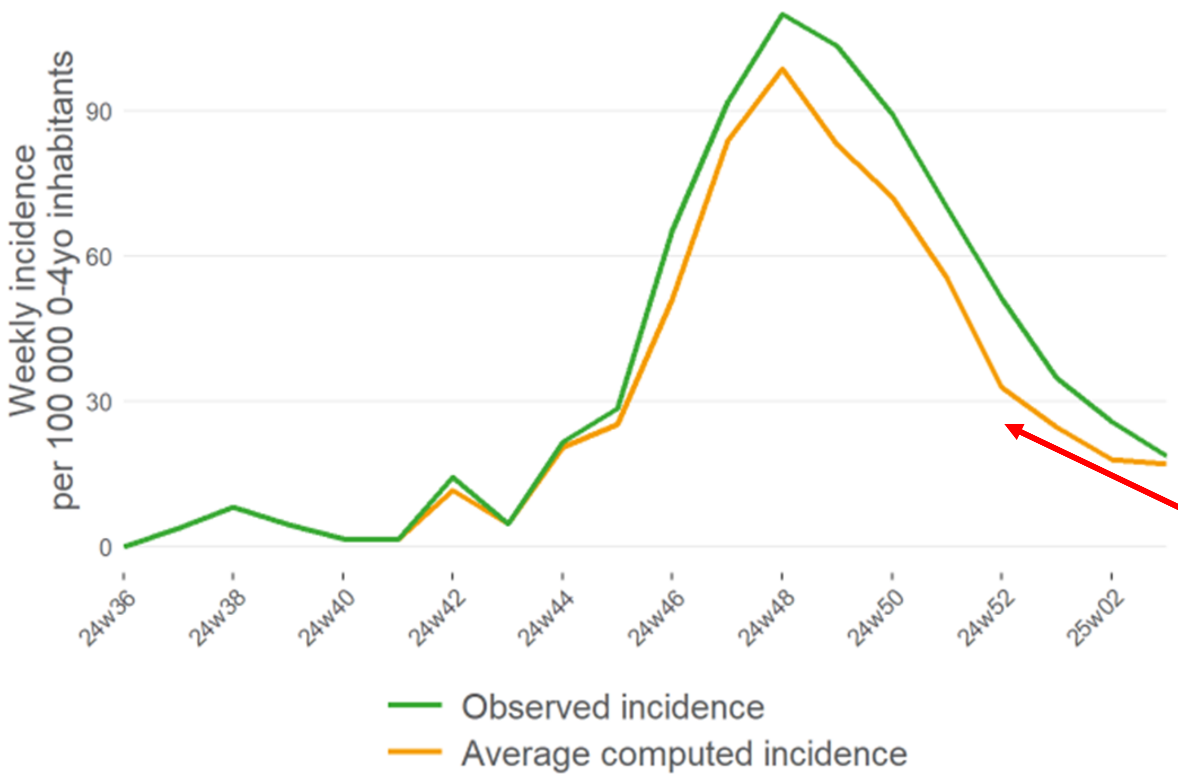
113 x

| Efficacy | |
|---|---|
| Low CI efficacy | 0.85 |
| Average efficacy | 0.88 |
| Hihg CI efficacy | 0.91 |

=

| Efficacy | Number of patients | Proportion of patients |
|---|---|---|
| Low CI efficacy | 96 | 20.7 |
| Average efficacy | 100 | 21.6 |
| Hihg CI efficacy | 103 | 22.2 |

# If all eligible patients would have received Nirsevimab (100% uptake) this season (Oct 2024-Janv 2025, SARI data)




**Incidence extrapolation:**

Before removing patients:

**749 per 100.000 0-4yo inhabitants**

After removing patients:

**618 per 100.000 0-4yo inhabitants**

Difference of:

**131 per 100.000 0-4yo inhabitants**

**→ extra reduction of**

**17.4 % of the incidence observed**

# Why was the catch-up uptake lower than expected??

*Acceptability of RSV prevention to Parents and Health Care Workers*

*Blumental S and Grossman D, Belgian Journal of Pediatrics Decembre 2024*

- **Multicenter opinion survey** among **young parents and antenatal caregivers** in Brussels

**→ New acceptability study in preparation**

**Broader recruitment of parents:**
Chirec Network, UZ Ghent, Jessa ziekenhuis + ONE, K&G and Kaleido

+ **Assessment of opinion of pediatricians, gynecologists and midwives all over the country**

**Goals:** identify expectancies toward prevention
+ **strengths and caveats** of the current campaign

# Take home messages
## Promising results but we can do better!


- From Oct to end December: **reduction of RSV pediatric hospitalizations** in 2024-2025 (first season with RSV prevention) compared to 2023-2024
- **Highest impact** seen among the **< 6months old** (in terms of number of cases and severity)
- **Reduction of RSV PICU burden**
- **Sub-optimal uptake** of the preventive tool in the catch-up cohort (better in maternity)

**Efforts should be done** to:

- Identify expectancies and feelings of parents and health care workers about RSV prevention
- Improve communication about the campaign through various channels
- Alleviate administrative workload linked to the preventive campaign

- **Participation of every Belgian center** (whatever its size, its region or its budget) is important to **monitor prevention impact on a national scale**

# Perspectives

- **In-depth analysis and comparison between both seasons** of pattern of RSV infection, antibiotics consumption, coinfections rate and complications.

- **Third year of study: some changes for the new upcoming season**:
  - Two tools reimbursed: nirsevimab and maternal vaccine

    Uptake and impact of both tools should be assessed together and separately
  - Expected better uptake

    Better communication, changes in reimbursement criteria, better awareness of parents and HCWs
  - Request submitted for reimbursement of nirsevimab for children with comorbidities

    (for their second RSV season, decision pending)

- **Cost effectiveness analysis from KCE ongoing**

- **Future new tools coming soon**

(new monoclonals : clesrovimab, vaccines for older children)

# Acknowledgements


- The team surveillance of respiratory infections (EID), Sciensano

Mathil Vandromme, Yves Lafort, Sven Hanoteaux, Izaak Van Evercooren, Yinthe Dockx, Koen Blot

- Danielle Strens (Realidad)
- Hilde Klechtermans (Jessa Ziekenhuis)
- Nathalie Bossuyt (Sciensano)

- The Care Fundation Chirec

# Thank you!!

Mélanie Devallée/ Parsy Marie-Pierre

Isabel Leroux-Roels/Levi Hoste

Xavier Holemans

Anne Tilmanne/Sandrine Roisin

Arne Witdouck/Siel Daelemans

Murielle Vanvaerenberg

Ivo Corthouts

Johan Hellinckx

Jan Vandersnickt

Celine Mignon

Wassim El Bittar

Katrien Coppens

Inge Matthijs

Tessa Goetghebuer

Kim Van Hoorenbeek

Amaury Delestienne

Alicia Wampers/Hilde Van Hauthem

Julie Frere/Catherine Sondag

Valbona Selimaj

Koen Vanlede

Sophie Van Speybroeck

Marc Bourgeois/David Tuerlinckx

Suzan Vansteijn/Caroline Van Rossem

Andrea Nebbioso

Jonas Dewulf

Tine Vanackere

Lore Winters

Anne-Sophie Bouteiller/Delphine Paige

Hanne Rigolle

Anne Tilmanne/Yoann Marechal

Marijke Reynders

Daan van Brusselen/Reinout Naesens

Sarah Dorval/Alexandra Van Eyck

Koen Magerman/Marc Raes

Olga Chatzis